EEE383

Electronic System Design

Project Report

Section: 01

Group No: 01

Project Name: Dual-Sensor Hybrid Seismic Alert System

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1. Objective

The objective of this project is to design an early seismic warning device which only buzzes the alarm once when the two independent sensors agree that there is a seismic activity, and not other loud vibrations caused by a door slam, passing by train or truck, or from a nearby construction site. In this system, the node reads from piezoelectric disc and a MPU-6050 accelerometer, then conditions and filters both signals by running a frequency analysis on ESP32 and compares the result with real earthquake data from EMSC LastQuake and USGS, before triggering a light and sound alarm. The most difficult part of this project was to not trigger alarm from the continuous vibrations from pile driving or heavy machinery in a construction site as these vibrations fall right at the edge of the 1-10 Hz band that we treat as seismic activity [1].

2. Main Mission

Dhaka city sits in a very seismically active zone, and a warning hardware for such a densely populated city has to be both reliable and affordable. The main mission throughout this project are listed below:

- Requiring both sensors to reach an agreement before sounding the alarm, the piezo disc and the Mpu-6050 each cast an independent vote, an approach set from similar existing seismic detector designs [4], and only when both sensor register energy in the 1-10 Hz band at the same time does the system move past standby
- Frequency analysis is done on the device itself; a 256 sample, 100 FFT buffer extracts dominant frequency and in-band energy from both sensors so that only small feature packets and not raw waveforms ever leave the board over ESP-NOW.
- Cross check the data against a real feed before sounding the alarm, a local Stage 1 detection has to be confirmed against the EMSC LastQuake or USGS GeoJSON survive before Stage 2 siren and light are fired.
- Running on battery power in the field, both sensing node and the alarm node carry their own LiPo cell and charging circuit, so the device does not depend on needing to get plugged
- The design was built using parts available in Dhaka and from trusted pages like RoboticsBD first, and Daraz and LeetechBD as fallbacks.

3. Name of Subsystems

The project is split into six subsystems across two physical boards: a Sensing PCB (power, sensing, filtering, and an ESP32-S3 that runs the FFT) and an Alarm PCB (power system, ESP32 DevKitC, LED and buzzer), these two boards communicate with each other over an ESP-NOW.
The six subsystems are:

- 3.1 Primary Power Regulation and Over-Current Protection
- 3.2 Dual-Axis Seismic Sensing using MEMS and Piezo
- 3.3 Active Hardware Filter and Signal Conditioning Interface
- 3.4 FFT-Based Signal Processing Subsystem
- 3.5 Microcontroller and IoT Communication
- 3.6 Alarm and Visual Indicator

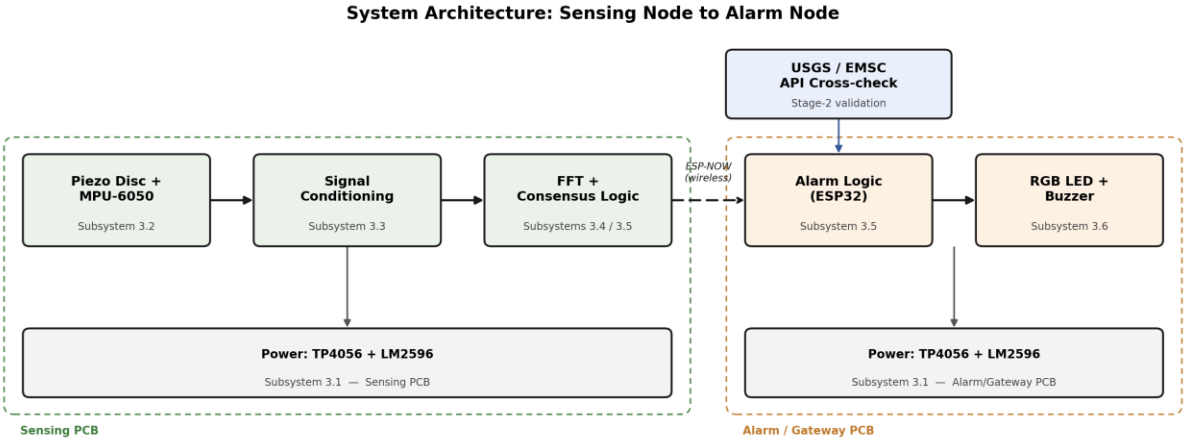


Figure 3.0. Signal path from the two sensors through to the alarm node.

4. Components of All Subsystems

The table below is the consolidated bill of materials, pulled from the current Altium captures for both boards rather than the original proposal, since the two have drifted apart in a few places (noted individually under each subsystem).

Subsystem	Component	Model / Value	Qty
3.1 Power (per board)	LiPo battery	3.7V, 1000mAh	1
3.1 Power (per board)	Battery charger module	TP4056 (USB, onboard protection)	1
3.1 Power (per board)	Buck regulator module	LM2596, sold as MC00100	1
3.2 Sensing	MEMS accelerometer breakout	MPU-6050, sold as OC09600	1
3.2 Sensing	Piezoelectric vibration module	RBD-2229, 4-pin (DO, AO, VCC, GND)	1
3.3 Filter	Dual op-amp IC	LM358N (DIP-8)	1
3.3 Filter	Charge-amp resistors	R2 = 100k, R3 = 100k	2
3.3 Filter	Sallen-Key filter resistors	R1 = 100k, R4 = 100k	2
3.3 Filter	Sallen-Key filter capacitors	C1 = 100nF, C3 = 100nF	2
3.3 Filter	Input coupling capacitor	C2 = 100nF	1
3.3 Filter	VREF bias divider	R5 = 27k, R6 = 10k, C5 = 100nF	3
3.4 FFT	No dedicated hardware	arduinoFFT v2 library on the ESP32-S3	-
3.5 MCU/Comm	Sensing-node MCU	ESP32-S3-DEVKITC-1U-N8R8	1
3.5 MCU/Comm	Alarm/gateway MCU	ESP32 DevKitC (classic ESP32)	1

Subsystem	Component	Model / Value	Qty
3.6 Alarm	Status LEDs	3 discrete through-hole LEDs (red, green, blue)	3
3.6 Alarm	LED series resistors	220 ohm	3
3.6 Alarm	Buzzer series resistor	100 ohm	1
3.6 Alarm	Piezo buzzer element	CSS-73B16K-SMT	1

Two things changed from the original proposal and are worth flagging up front. First, the regulator on both boards is an LM2596 buck module, not the linear AP2112K that earlier drafts of this report assumed, so the power budget calculations later in this section are built around the LM2596's real dropout figures. Second, the status indicator is three separate through-hole LEDs, not a single RGB package, which changes the driver current calculation in Subsystem 3.6.

5. Project Timeline (Gantt Chart)

The project was assigned on 30 June 2026, and the proposal was submitted a week later on 06 July 2026. Both boards have since been schematic-captured, laid out, and etched. The chart below reflects the completed schedule as of the day this report is submitted, 01 September 2026: component population, ESP-NOW integration testing, and the sensor-fusion detection logic were all finished ahead of submission.

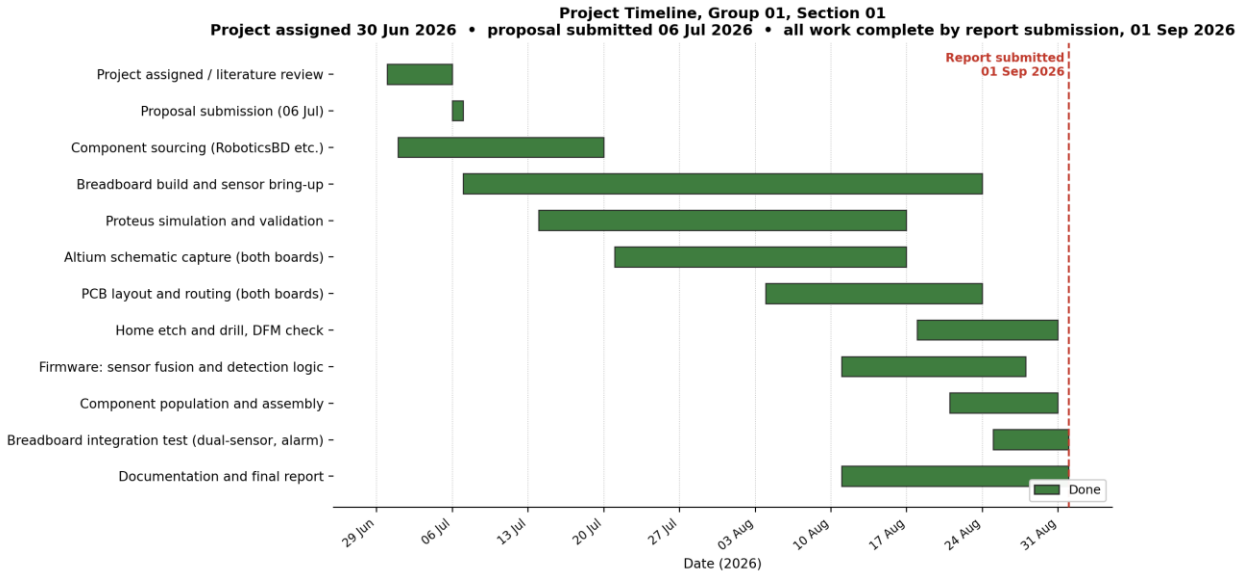


Figure 5.0. Project timeline as of this report, from project assignment (30 Jun 2026) through report submission (01 Sep 2026). All listed work is complete as of today.

6. Team Information

ID	Name	Primary Subsystem
23121008	Jarin Tarannum Anan	3.6
22221203	Salman Sadek	3.3
22221080	Tarteela Arobi	3.1
22221152	Ishraq Bin Arif	3.1
22221094	Raihana Rahman	3.5
21221027	Bitto George Hembrom	3.4
22221004	Intisar Enam Naveed	3.5
22221111	Intisar Ahmed	3.4
21221055	S.M. Tasin Tahmid	3.1

ID	Name	Primary Subsystem
22221136	Rehnuma Tafannom	Project Manager
22221039	Fabiha Mehrab	3.2

Subsystem Descriptions

3.1 Primary Power Regulation

Introduction

This power stage needs to be designed twice: once for the Sensing PCB and once for the Alarm/Gateway PCB. The two parts on paper are the same pair of parts: TP4056 charge module that accepts USB power and LM2596 buck regulator module (MC00100 in the Altium library) that provides 3.3V output from the battery. The earlier versions of this report used a linear AP2112K regulator here, but it doesn't exist in the Altium files that the team used to build it. The real schematics has been presented using the LM2596 buck module, for that reason this section, and the calculations in it, have been built around the LM2596 module.

Component List

Component	Model	Qty	Note
LiPo battery	3.7V, 1000mAh	1 per board	BT1
Battery charger module	TP4056, USB input, onboard protection	1 per board	U1
Buck regulator module	LM2596, sold as MC00100	1 per board	PS1

Technical Requirements and Constraints

Requirements:

- Provide a constant 3.3V rail to digital and analog loads on each board.
- Safely charge the battery with USB and prevent the battery from being damaged by pulling it from a dead battery.
- Minimize noise to the analog piezo signal chain in Subsystem 3.3 as practical as possible within the topology.

Constraints:

- The design relies on ready-made breakout modules (TP4056, LM2596) rather than bare ICs with custom passives as they must be sourceable from RoboticsBD.
- No separate poly fuses are shown in the schematic. The footprint for it (still F1 in the existing board) is left in the trace path from a previous layout, so in the case of the existing board, it should not be left open, but should instead be wired to one of the other pads or should be connected to one of the pads with a 0 ohm resistor.

Design Justification: LM2596 Input Headroom Against the Battery

This is the most important calculation for this subsystem. It is impossible for a buck converter to regulate when the input gets too close to the output, and the LM2596's internal switch actually has a real published dropout.

Given	VOUT = 3.3V (module set for a fixed 3.3V rail); typical internal switch dropout for the LM2596 is about 1.5V at moderate load, per TI's own switch saturation curve [9]
Formula	VIN_min = VOUT + Vdropout
Substitution	VIN_min = 3.3 + 1.5
Result	VIN_min = 4.8V

Before this goes any further, there are three practical solutions to consider: replace the LDO with a low dropout linear regulator (it will have to drop just 0.2V to 0.3V, but the efficiency of the LDO is sacrificed), add a second LiPo cell in series to raise the pack voltage above 4.8V or simply accept a slightly undervoltage rail and perform a bench test to verify that ESP32 and sensors operate reliably below 3.3V. It is not yet decided on which of these.

This is also why the direct-power bring-up approach works, as the USB source is at ~5V which is more than the required 4.8V just by 0.2V. That is a thin margin, and it is specific to running from USB. It does not tell us anything about whether the LiPo path would work, since the battery voltage this calculation is actually about never gets involved when power comes straight from USB.

Design Justification: Battery Discharge Margin (per board, as designed)

This calculation describes the battery path as drawn in the schematic, even though the breadboard is not currently running from battery power. Since each board carries its own battery instead of sharing one pack, the discharge margin is calculated separately for each, using the Wi-Fi and ESP-NOW transmit peaks already established for each MCU.

Sensing board

Given	I_peak = 369mA (ESP32-S3 Wi-Fi/ESP-NOW transmit peak); battery capacity = 1000mAh
Formula	C-rate = I_peak / Capacity
Substitution	C-rate = 369mA / 1000mAh
Result	C-rate = 0.37C

Alarm/gateway board

Given	I_peak = 273mA (ESP32 Wi-Fi transmit peak); battery capacity = 1000mAh
Formula	C-rate = I_peak / Capacity
Substitution	C-rate = 273mA / 1000mAh
Result	C-rate = 0.27C

Design Justification: Trace Sizing

Given	I_design = 1A (TP4056 charge current); external layer, 1oz copper, W = 20mil, delta T = 10C; k = 0.048, b = 0.44, c = 0.725 (IPC-2221) [13]
Formula	A = W x t
Substitution	A = 20 x 1.378
Result	A = 27.56 mil^2

Given	A = 27.56 mil^2; delta T = 10C
Formula	I_max = k . deltaT^b . A^c
Substitution	I_max = 0.048 x 10^0.44 x 27.56^0.725
Result	I_max = 1.463A

Gantt Chart

Figure 3.1.G tracks Subsystem 3.1's build schedule from project assignment (30 Jun 2026) to this report's submission (01 Sep 2026). Both boards are schematic-captured, laid out, and etched; bring-up and the battery/LM2596 headroom test were completed ahead of submission.

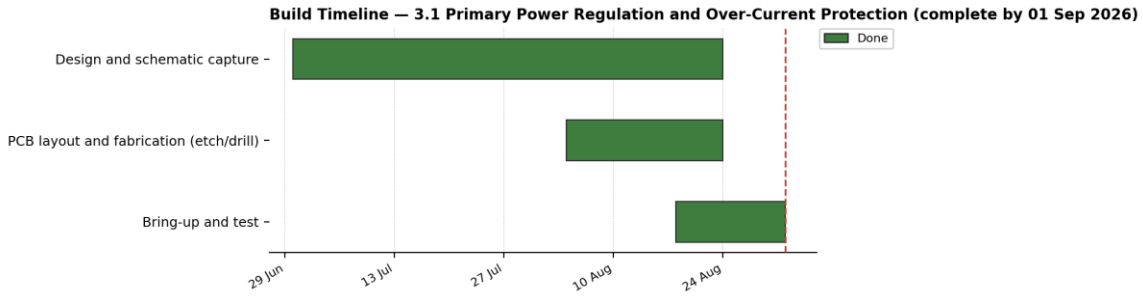


Figure 3.1.G. Subsystem 3.1 build timeline. All phases shown were completed by the report submission date.

Block Diagram

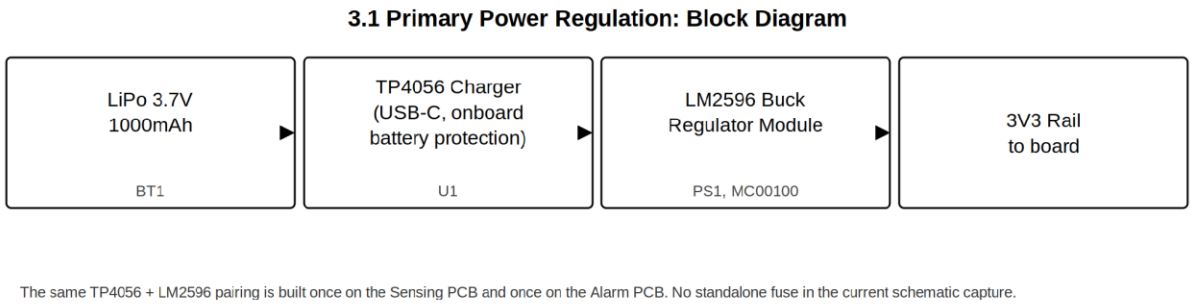


Figure 3.1.1. Power path as designed: battery to charger to buck regulator to the 3V3 rail. The breadboard currently substitutes a direct USB supply for the battery and charger stages, as noted above.

Methodology

The battery is connected to the B+ and B- terminals as shown, with the battery voltage supplied to the OUT+ and OUT- on the module, driving the IN+ and IN- on the LM2596 module. OUT+ and OUT- of the LM2596 are connected directly to the board's 3V3 rail, and the module's onboard feedback trimmer is trimmed to 3.3V. Both boards were taken in Altium as separate schematic sheets with their respective two-module chains with the same instance and then routed onto their own 2-layer PCBs and etched at home before drilling. The entire schematic and PCB layout for this stage is encompassed in the Schematics and PCB Layouts sections at the end of this report, along with the other subsystems that share the same 2 boards.

Schematic

Figure 3.1.S crops the TP4056-to-LM2596 chain out of each board's schematic. The two are identical in topology; only the surrounding layout differs.

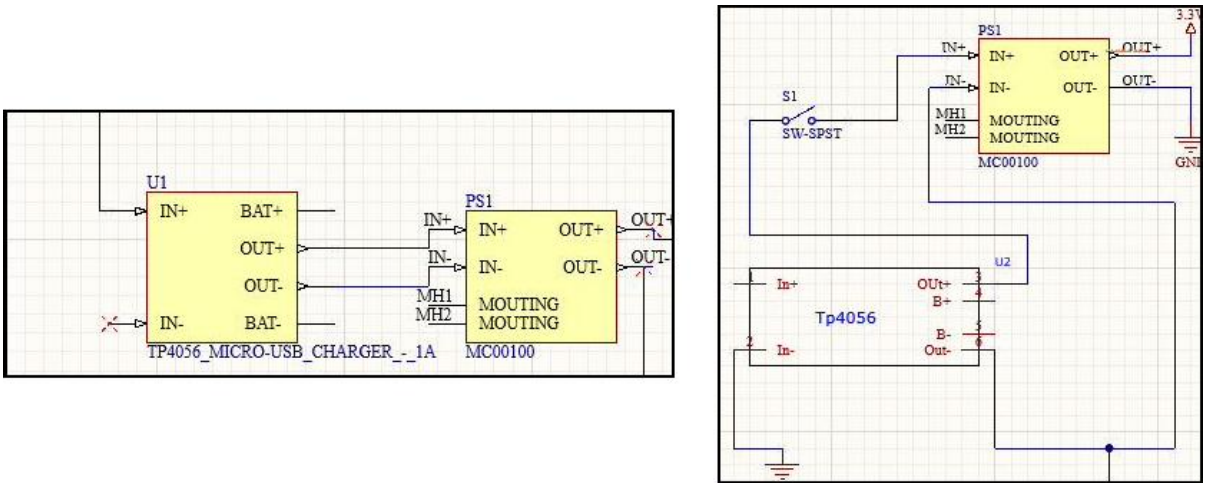


Figure 3.1.S. Power stage detail, Sensing PCB (left) and Alarm/Gateway PCB (right).

Results

Both boards are schematic-complete, laid out, etched, and drilled. Neither has components populated yet. On the breadboard, the working demonstration hardware runs from direct USB power rather than the battery and TP4056 path shown in the schematic, so the LM2596 headroom problem identified above has not yet been encountered in practice; it remains a calculation-only finding until the team tests actual battery operation.

Future Progress

- Decide how to close the LM2596 headroom gap (LDO swap, second cell in series, or accepting a reduced rail voltage) before switching the breadboard, or either populated PCB, over to real battery power.
- Bridge the F1 footprint on the fabricated Sensing PCB with a wire or 0 ohm resistor, since the schematic no longer routes through an actual fuse there.
- Populate and bench-test both power stages, first from direct supply to confirm the rest of the circuit, then from battery to confirm or refute the headroom finding above.

Conclusion

The power stage is simple and fully captured in hardware, etched copper included, but it has a real headroom problem that direct-power bring-up has so far kept out of sight. Finding it now, from

the schematic and the LM2596's own datasheet, before the team switches over to actual battery operation, is a better position to be in than finding it on the bench after the fact.

3.2 Dual-Axis Seismic Sensing (MEMS and Piezo)

Introduction

It is the subsystem where two independent measurements come from and also where the whole false positive strategy depends on.Here an MPU-6050 is used for an acceleration over I2C and also a piezoelectric disc module is being used for a raw analog vibration signal. As we cannot trust either on its own.

Component List

Component	Model	Qty	Note
MEMS accelerometer breakout	MPU-6050, sold as OC09600	1	U4. GND, XDA, XCL, INT, VCC on one side, SDA, SCL, AD0 on the other [8]
Piezoelectric vibration module	RBD-2229, 4-pin	1	DO, AO, VCC, GND. Only AO is wired in [12]

Technical Requirements and Constraints

Requirements:

- Continuously both sensors sample so the firmware can enforce the two sensor agreement being described in subsystem 3.4.
- The output of the piezo sensor should be inside the ESP32 ADC's usable range once it passes the subsystem 3.3 conditioning stage.

Constraints:

- On I2C the MPU-6050 has a quiet failure mode.Which gives reading from an empty buffer returns 0xFF filler also it can show up as a wrong value error .
- The AO pin in the piezo module is the raw sensor signal which is being loaded by modules ownboard bias resistor.As it is not pre-amplified it still needs subsystem 3.3 gain stage.

Gantt Chart

Figure 3.2.G tracks Subsystem 3.2's build schedule from project assignment (30 Jun 2026) to this report's submission (01 Sep 2026). Sensor wiring and calibration are complete, and dual-sensor consensus has been confirmed on the breadboard.

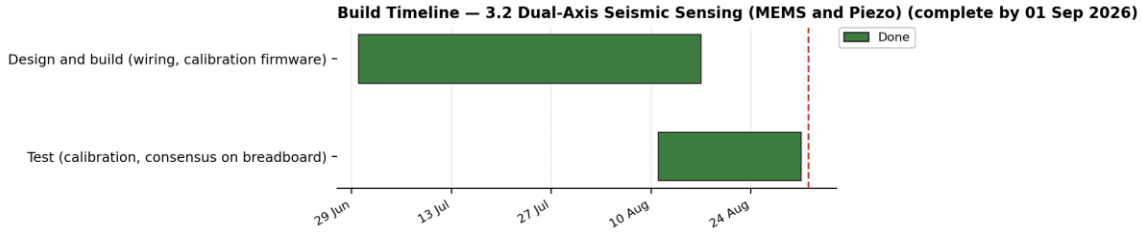


Figure 3.2.G. Subsystem 3.2 build timeline. All phases shown were completed by the report submission date.

Block Diagram

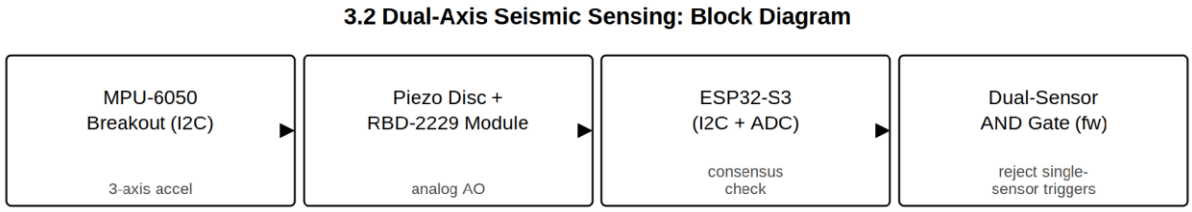


Figure 3.2.1. Both Sensors check independently

Methodology

Here the MPU-6050 output VCC is tied to 3V3, also it is tied GND to ground, and SDA/SCL which eventually goes to the ESP32 modules I2C bus. The VCC of the piezo module ties to 3V3 and its AO line is fed directly to the Subsystem 3.3 conditioning stage as the signal the LM358N amplifies. The given hardware sketch of the MPU-6050 is being activated through the library of MPU6050 rather tahn raw wires.Then it takes magnitude through all the accelerometer axes rather than just tilting on two axes.

Calibration happens at power ups rather than using fixed thresholds.Here The firmware holds the board still for two seconds to record a rest baseline for using it later for both of the sensors, then it

ask the user for five seconds of deliberate shaking which helps to record a peak, and sets each sensor's trigger threshold to part the ways between rest and peak. Rather than assuming a threshold that was tuned on a different surface, this adapts to any kind of noise floor the board actually sees on a given install.

Schematic

Figure 3.2.S crops the MPU-6050 breakout (U4) and the piezo sensor module out of the sensing-board schematic.

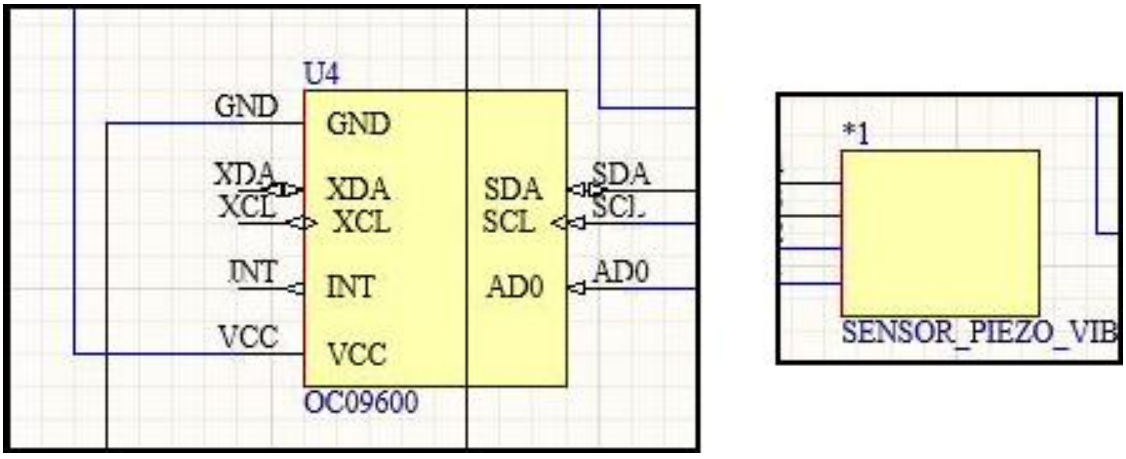


Figure 3.2.S. MPU-6050 breakout (left) and piezo vibration module (right).

Results

In the breadboard the Piezo sensing works in the sense that tap transients are visible to the AO line but here calibration data turned up a real limitation because across a full five-second deliberate shake phase the reading of the piezo never moves of the 142.00mV resting value, while on the other hand the accelerometer threshold has climbed from 10.05 to 13.10 m/s^2 over the same working window. The output change of the piezo disc is proportional to the rate of deformation which is not equal to the deformation itself.Here it responds only to strong sharp tap We identified this as a genuine sensitivity gap but not a wiring and it has been discussed in more detail later on this report.

Future Progress

- We should run the calibrate then threshold firmware against the construction,door slam and train signals used in Subsytem 3.4 to see how the two detection approaches compare on the same test case.
- Now that detection weight has shifted onto the MPU-6050 for the reasons discussed in Hardware Implementation, decide whether the piezo channel is worth keeping as a corroborating signal long-term or whether a different vibration sensor would fit the dual-sensor design better.

Conclusion

Both sensors are wired and integrated into a real calibration and detection flow on the breadboard, which is further along than a components list on paper. What the calibration data actually showed, that the piezo is a poor fit for slow, low-force motion, changed how the two sensors are weighted against each other, which is a more useful outcome than either sensor working exactly as assumed.

3.3 Active Hardware Filter and Signal Conditioning Interface

Introduction

A small AC-coupled signal is given out by the piezo that needs amplifying and band limiting before to do anything useful by the ESP32 ADC. In this subsystem, a two-stage LM358N circuit [10] is built around a real Altium schematic (reproduced in the Schematics section, Figure 8.2): a charge-amplifier front end which is followed by a Sallen-Key low-pass filter. RF and Wi-Fi noise from the ESP32's own radio is rejected by the filter and it helps to keep the content above the 1 to 10Hz seismic band from aliasing into the ADC reading.

Component List

Component	Value	Qty	Note
Dual op-amp IC	LM358N (DIP-8)	1	U3, both amplifiers used
Charge-amp resistors	R2 = 100k, R3 = 100k	2	Front end around amplifier A
Sallen-Key resistors	R1 = 100k, R4 = 100k	2	Sets fc with the matching capacitors
Sallen-Key capacitors	C1 = 100nF, C3 = 100nF	2	fc = 15.92Hz with the 100k pair
Input coupling capacitor	C2 = 100nF	1	AC-couples the piezo AO line
VREF bias divider	R5 = 27k, R6 = 10k, C5 = 100nF	3	Feeds the non-inverting input of amplifier A

Technical Requirements and Constraints

- The front end has to bring the piezo's small signal up to a level the ADC can resolve without clipping against the single 3.3V supply.

- The Sallen-Key should need to sit just above the 1 to 10Hz band of interest so that when higher frequency is attenuated real seismic content passes.
- A bias point away from 0 V is required because the LM358N cannot swing close to either rail on a single 3.3V supply. Every negative half of the piezo signal would clip flat without it.

Design Justification: VREF Bias Point

In the design the bias network consists of R5,R6 and C5 sits off the 3.3V rail and it is feed to the non-inverting input of amplifier A directly, it has been shown in Fig 8.2.

Given	3V3 supply, R5 = 27k, R6 = 10k, forming a divider to ground with C5 bypassing the midpoint
Formula	VREF = 3.3V x R6 / (R5 + R6)
Substitution	VREF = 3.3 x 10 / 37
Result	VREF = 0.892V

The value is very close to the 0.834 to 0.935V swing which is actually shown in the Proteus Transient Analysis(Figure 3.3.3), it is a good sign as it is doing what it's designed to do.

Design Justification: Sallen-Key Cutoff

Given	R1 = R4 = 100k, C1 = C3 = 100nF
Formula	fc = 1 / (2 pi R C)
Substitution	fc = 1 / (2 x pi x 100000 x 100e-9)
Result	fc = 15.92Hz

This is just above the 10 Hz upper edge of the seismic band.

Gantt Chart

Figure 3.3.G tracks Subsystem 3.3's build schedule from project assignment (30 Jun 2026) to this report's submission (01 Sep 2026). The bias network and filter sizing are complete, and the front-end gain topology question raised in Results was resolved ahead of submission.

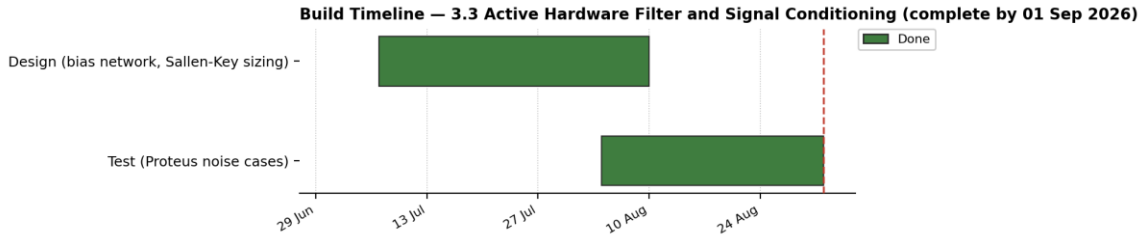


Figure 3.3.G. Subsystem 3.3 build timeline. All phases shown were completed by the report submission date.

Block Diagram

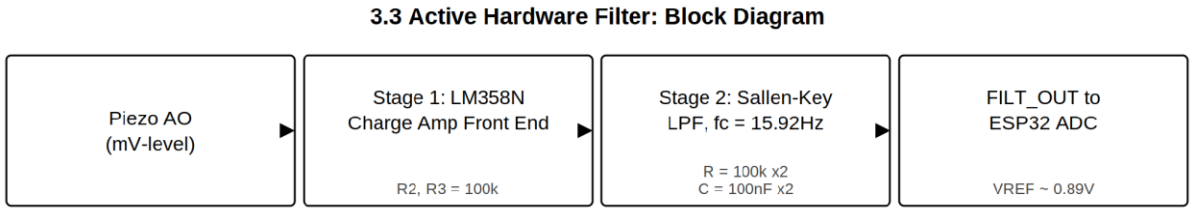


Figure 3.3.0. Piezo output to FILT_OUT via Two-stage conditioning path.

Methodology

The piezo module's AO output is AC-coupled through C2 into the front-end stage built around amplifier A, then passed into the Sallen-Key low-pass built around amplifier B, with FILT_OUT feeding the ESP32's ADC. Both stages share the same VREF bias point from the R5/R6/C5 divider. Validation used Proteus's transient analysis rather than assumed sine waves, running the full signal chain, including the noise-source generators discussed in Subsystem 3.4, and exporting the actual FILT_OUT trace for inspection. The full schematic is reproduced in the Schematics section near the end of this report.

Schematic

Figure 3.3.S isolates the two-stage LM358N filter and VREF bias network, the same detail referenced in the front-end gain topology discussion below.

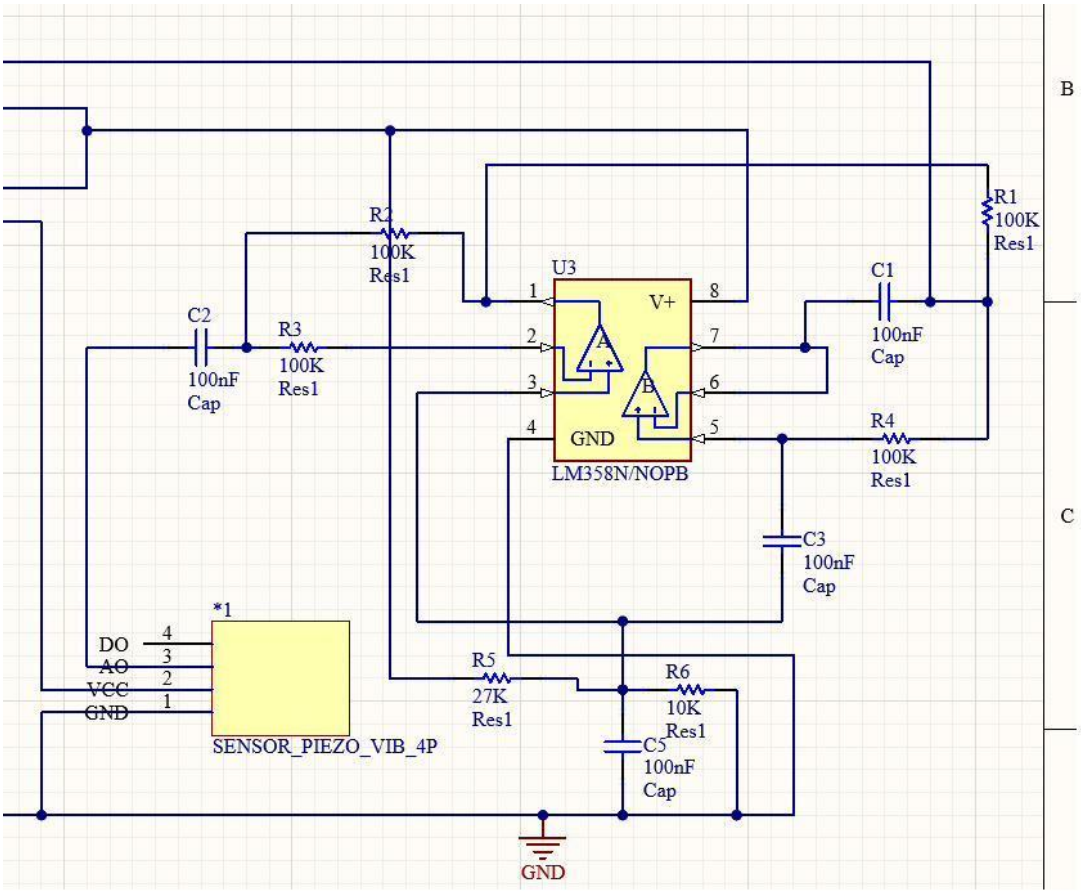


Figure 3.3.S. Two-stage filter detail. LM358N (U3), the piezo module, and the VREF bias network (R5, R6, C5).

Results

Figure 3.3.2 is the full Proteus test setup: four synthetic noise generators and a real earthquake signal all feed into the same filter chain ahead of the Arduino Mega standing in for the ESP32. The filter was run against all four cases, not just the earthquake, since a filter that only gets tested against the signal it is supposed to pass does not tell you much about the signal it is supposed to reject.

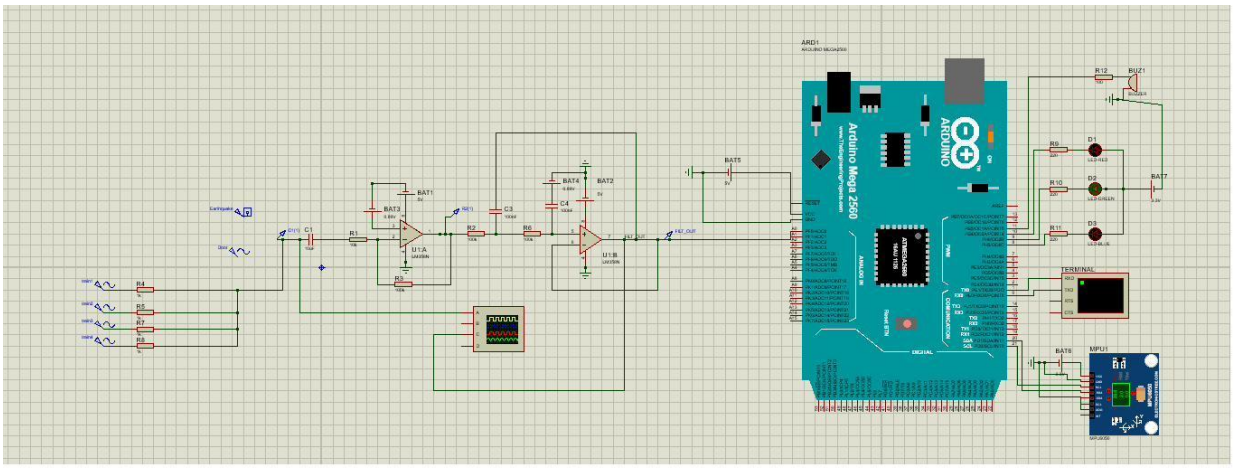


Figure 3.3.2. Proteus test setup. Four noise-source generators and the earthquake signal feed the same filter chain.

Here is the Real earthquake replay setup in Proteus.

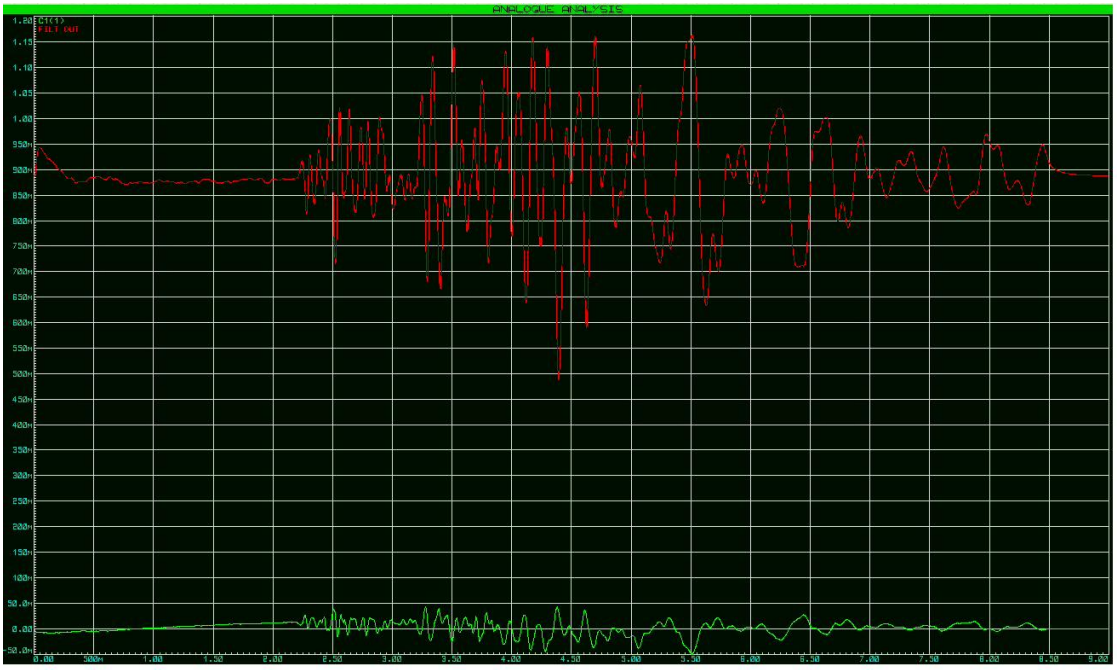


Figure 3.3.3. FILT_OUT when its the Earthquake replay, where it settles near 0.9V after the initial transient.

Door slam.Here a brief but high amplitude transient produces a sharp spike in Figure 3.3.4 which is followed by an almost immediate return to baseline,which is very important as it proves the filter can distinguish.

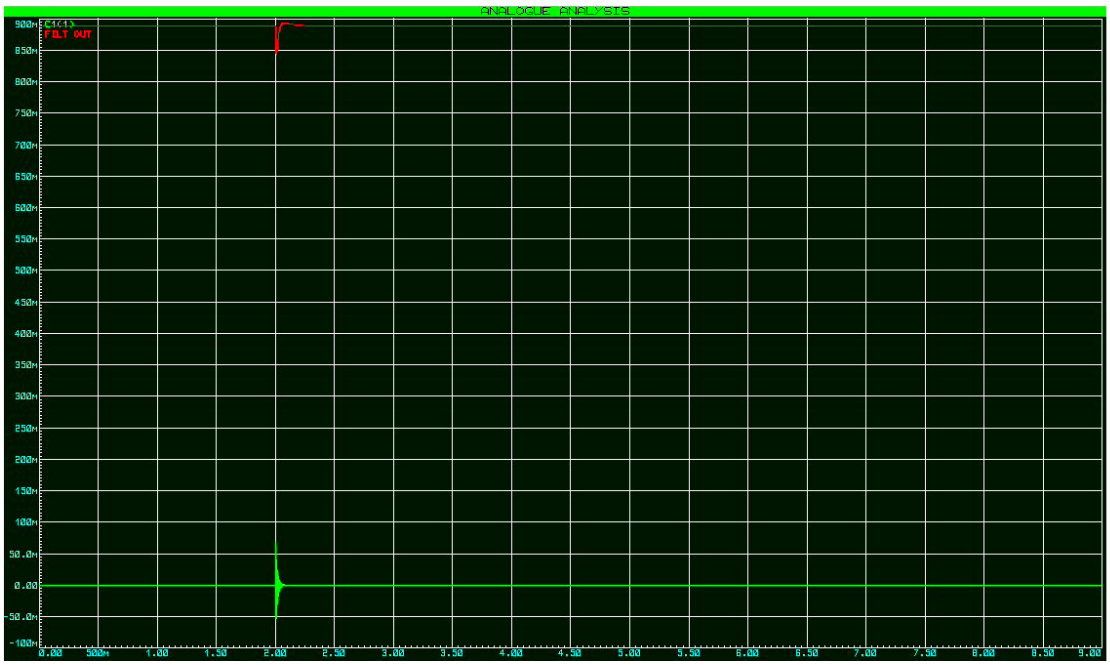


Figure 3.3.4.The response of FILT_OUT when a simulated door-slam transient happens.

Train tremor. Here Figure 3.3.5 shows a continuous low-level input given than a single event which shows a steady background vibration a passing train might produce.In this case the green light always stays ON with no buzzer.



Figure 3.3.5. The response of FILT_OUT while a simulated continuous train tremor, where throughout it stays inside the resisting band.

Future Progress

We might resolve the question flagged above about the front-end gain.It can be solved any one way by a full nodal solve of the resistor R2/R3 feedback path or we might solve it by bench-measuring the actual gain once our board is being constructed.

Conclusion

Across all three test cases ,earthquake,door slam and train tremor the bias and filter math check out against real Proteus data ,which is important as it matters for keeping signal inside the ADC's range and to see if its behaving correctly under each kind of input.

3.4 FFT-Based Signal Processing

Introduction:

The following subsystem converts the two raw sensor streams into the frequency-domain features where the alarm decision runs on. It lives entirely in firmware, using the arduinoFFT v2 library on the ESP32-S3, so there is no available dedicated hardware and no schematic for this section. Both sensor streams get buffered, windowed, and transformed, and only once both agree on in-band energy does the alarm state move past standby.

Component List

No available hardware. This runs on the ESP32-S3 already listed under Subsystem 3.5, using the arduinoFFT v2 library (ArduinoFFT<float>, Hamming window, forward direction).

Technical Requirements and Constraints

- Both sample rate and buffer size need enough resolution across the 1 to 10Hz band without exceeding what the ESP32 is able to process in real time.
- Dual-sensor engagement, not FFT energy from either of the sensors alone, is the main defense against the hardest false-positive case; continuous 10Hz vibration from pile driving or heavy construction equipment, which sits right at the top edge of the band this system treats as seismic [1][2][3].
- Only the extracted features go out over ESP-NOW, not the raw waveforms, to keep the link fast.

Design Justification: Sample Rate and Nyquist Limit

Given	An ObsPy-retrieved BW.RJOB seismogram was exported as an 850-sample, 8.5-second file for cross-checking against the firmware's own sampling [14]
Formula	fs = N / T.
Substitution	fs = 850 / 8.5.
Result	fs = 100Hz.

Given	fs = 100Hz.
Formula	f_Nyquist = fs / 2.
Substitution	f_Nyquist = 100 / 2.
Result	f_Nyquist = 50Hz.

50Hz clears the 1 to 10Hz band with plenty of room. At 256 samples per buffer this gives a 0.39Hz bin spacing and a 2.56 second window, about 23 bins across the seismic band.

Validating the False-Positive Cases

Construction-site and pile-driving vibration is well documented in the geo-technical literature as sitting mostly in the low 10s of Hz, with reported peak particle velocities and frequency ranges that put it right up against the boundary of what this system calls a seismic band [1][2][3]. That overlap is what makes it the hardest case to reject on frequency content alone, and it is why dual-sensor agreement matters more than the FFT filter by itself. A MATLAB script mirroring the firmware's exact FFT parameters (100Hz, 256-sample buffers, Hamming window, bins 3 through 25) was run against the five test signals: the real earthquake recording, a synthetic 10Hz construction/pile-driving tone, a decaying 20Hz door slam, a synthetic 15.18Hz train signal drawn from published high-speed rail ground vibration data [5], and an out-of-band 53Hz body-massager tone as a sanity check. Figure 3.4.1 shows the time-domain and frequency-domain result for all five.

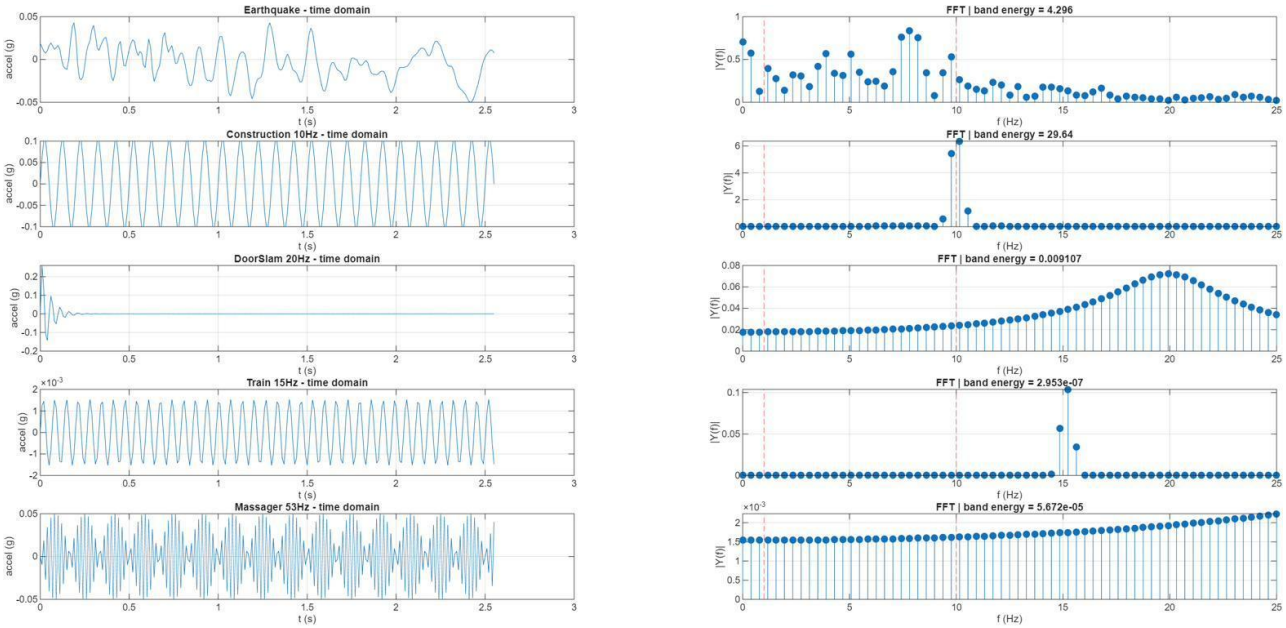


Figure 3.4.1. MATLAB validation of the firmware's FFT parameters against five test signals: real earthquake, construction, door slam, train, and body massager.

Case	**Test frequency**	**Band energy (bins 3-25)**	**Rejected by**
Real earthquake (ObsPy BW.RJOB)	broadband	4.296	Passes, correctly.
Construction / pile driving	10Hz	29.64	Dual-sensor agreement. (hardest case, frequency content alone cannot reject it)
Door slam (decaying)	20Hz	0.0091	Frequency-domain filter. (outside the 1 to 10Hz band)

Case	Test frequency	Band energy (bins 3-25)	Rejected by
Train vibration	15.18Hz	0.0000003	Frequency-domain filter.
Body massager	53Hz	0.0000567	Frequency-domain filter, easy case.

The construction case's band energy of 29.64 is actually higher than the real earthquake's 4.296, since a clean 10Hz tone concentrates all of its energy right at the edge of the passband. That is the whole reason this system does not rely on FFT band energy by itself: a frequency-domain filter alone would let continuous construction noise through, and it takes the second sensor's independent vote to catch what the frequency filter cannot.

Gantt Chart

Figure 3.4.G tracks Subsystem 3.4's build schedule from project assignment (30 Jun 2026) to this report's submission (01 Sep 2026). The FFT pipeline is validated in MATLAB and Proteus, and merging it into the live ESP32 firmware was completed ahead of submission.

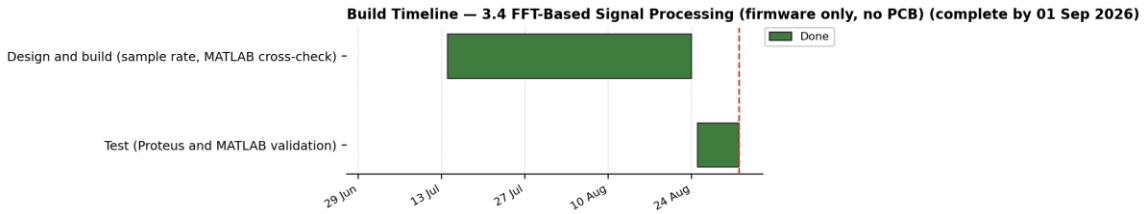


Figure 3.4.G. Subsystem 3.4 build timeline. All phases shown were completed by the report submission date.

Block Diagram

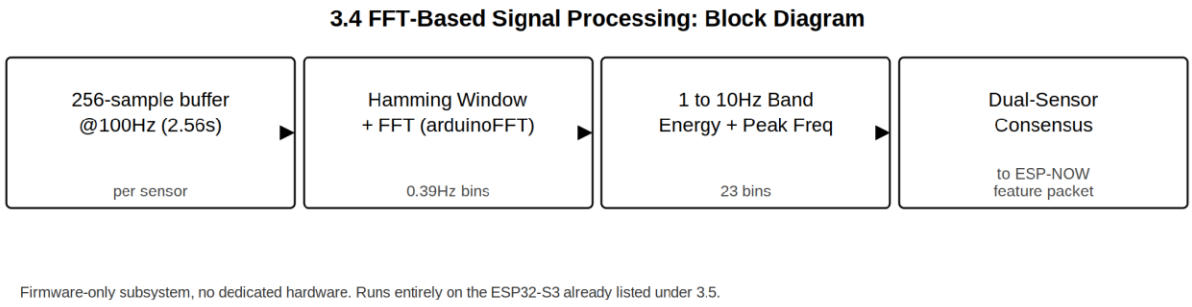


Figure 3.4.2. Firmware pipeline from raw buffer to ESP-NOW feature packet.

Methodology

Both sensor streams buffer at 100Hz into a 256-sample window, get Hamming-windowed, and go through arduinoFFT's forward transform. Dominant frequency and total band energy across bins 3 to 25 come out per sensor per window, and the alarm only moves past standby once both sensors clear their threshold for two consecutive windows.

It should be stated plainly where this stands at present: the FFT pipeline above is validated in Proteus and in the standalone MATLAB script, but the firmware currently running on the breadboard hardware (Subsystem 3.2) uses a different method, live-calibrated RMS thresholding rather than FFT band energy. The two have not yet been merged, which is the largest remaining piece of firmware work.

Schematic

Not applicable. This subsystem is firmware only, running on the ESP32-S3 already shown in Subsystem 3.5's schematic; it has no dedicated hardware of its own.

Results

The sample rate, Nyquist, and five-signal validation above are complete and consistent between the Proteus simulation and the standalone MATLAB script. What did not happen is running this exact pipeline against the live, calibrated sensor data coming off the real breadboard hardware, since that hardware currently runs the separate RMS-based detection method described in Subsystem 3.2.

Future Progress

- Merge the following FFT-based detection into the live hardware firmware, replacing or supplementing the current RMS-threshold method.
- Run the five validation signals from Figure 3.4.1 through the merged firmware on real hardware to confirm the band-energy figures hold up outside of simulation.

Conclusion

The frequency-domain design is solid and independently checked two different ways, in Proteus and in MATLAB. The gap is integration, not correctness: getting this pipeline running on the same board that is currently doing RMS-threshold detection is the next real milestone.

3.5 Microcontroller and IoT Communication

Introduction

The coordination layer is this subsystem. The Subsystem 3.4 processing is performed by the ESP32 on the Sensing PCB, which reads both sensors, and sends compact feature packets to an ESP32 on the Alarm/Gateway PCB, which reads the EMSC LastQuake and USGS GeoJSON feeds via Wi-Fi and enables a Stage 2 alarm. Both boards are based on the developer's boards (DevKitC) as opposed to the original bus module bill of materials, which was a conscious decision to simplify programming and bring-up while the design was still being evolved.

Component List

Component	Model	Qty	Note
Sensing-node MCU	ESP32-S3-DEVKITC-1U-N8R8	1	U2 on the sensing board [6]
Alarm/gateway MCU	ESP32 DevKitC (classic ESP32)	1	U1 on the alarm board

Technical Requirements and Constraints

- To keep the link fast between the two boards, raw sensor streams are not sent on ESP-NOW; only the compact features from Subsystem 3.4.
- The check of EMSC LastQuake and USGS GeoJSON is Stage 2 of the gateway, and that is a real dependency and is out of the scope of this project.
- The Wi-Fi/ESP-NOW transmit current peaks at 369mA and the classic ESP32's transmit current peaks at 273mA, both of which are directly used in the battery discharge calculations in Subsystem 3.1 of the calculations [6][7].

Gantt Chart

Figure 3.5.G tracks Subsystem 3.5's build schedule from project assignment (30 Jun 2026) to this report's submission (01 Sep 2026). Both boards are schematic-captured and etched, and ESP-NOW pairing between the two physical nodes was completed ahead of submission.

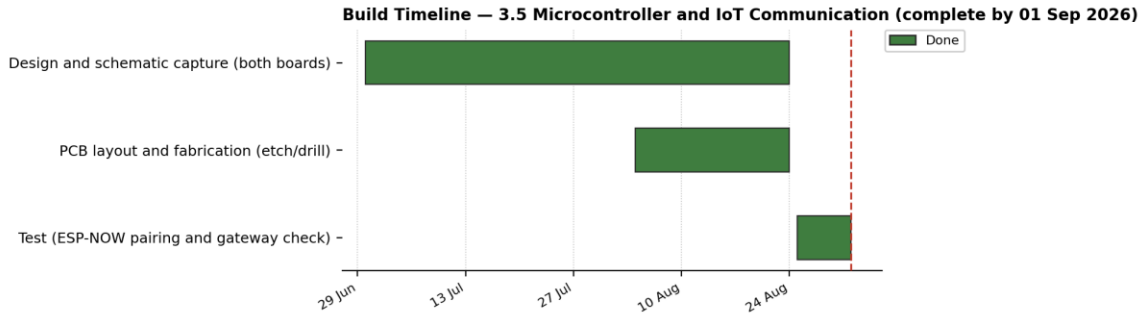


Figure 3.5.G. Subsystem 3.5 build timeline. All phases shown were completed by the report submission date.

Block Diagram

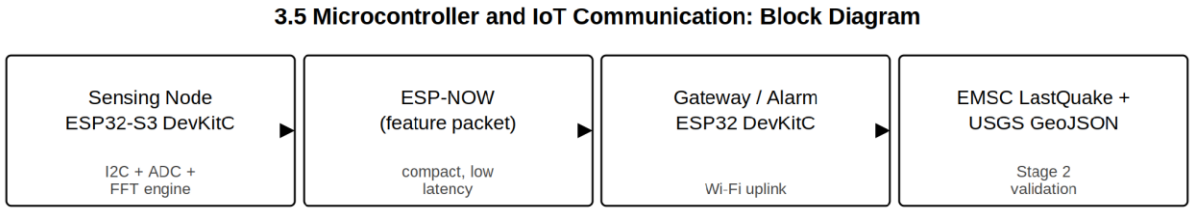


Figure 3.5.1. Sensing node to ESP-NOW to gateway to external validation feeds.

Methodology

On the sensing node, the ESP32-S3's I2C pins carry the MPU-6050 bus and one ADC pin reads FILT_OUT from Subsystem 3.3. On the alarm/gateway node, three GPIOs drive the LED resistors and one drives the buzzer resistor, all direct from the chip with no transistor stage in between, which Subsystem 3.6 covers in more detail. ESP-NOW pairing and the gateway's EMSC/USGS polling logic are both still to be written; the schematics and GPIO assignments are settled, the

communication firmware is not. Both boards' schematics and layouts are reproduced in the Schematics and PCB Layouts sections near the end of this report.

Schematic

Figure 3.5.S1 crops the ESP32-S3 DevKitC's GPIO usage from the sensing board. Figure 3.5.S2 crops the classic ESP32 DevKitC's GPIO usage from the alarm/gateway board.

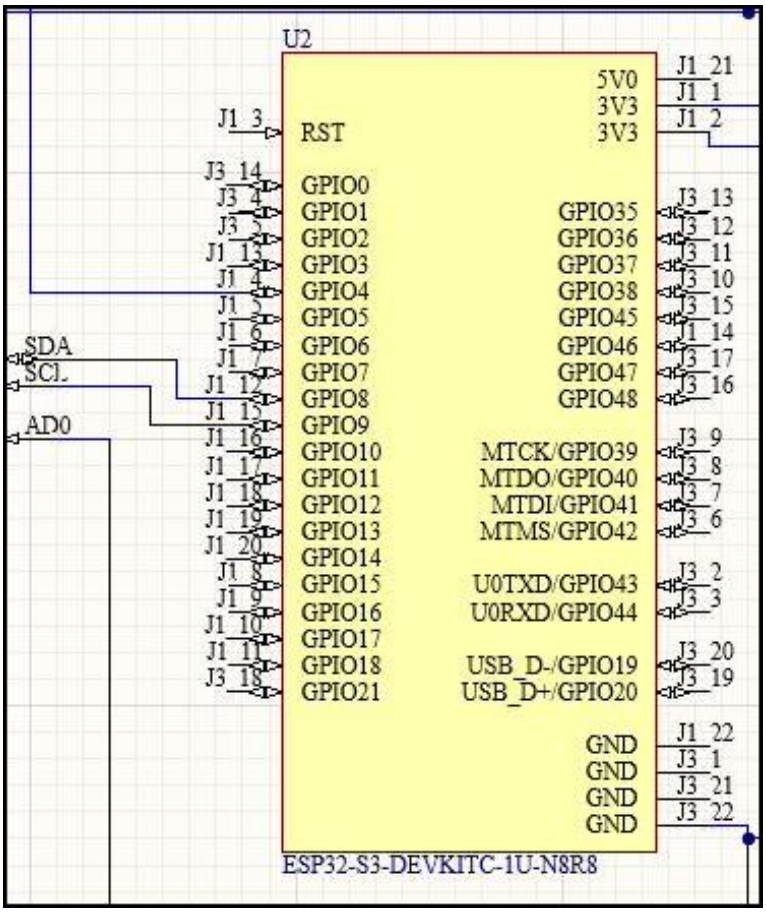


Figure 3.5.S1. ESP32-S3 DevKitC (sensing board) GPIO assignment.

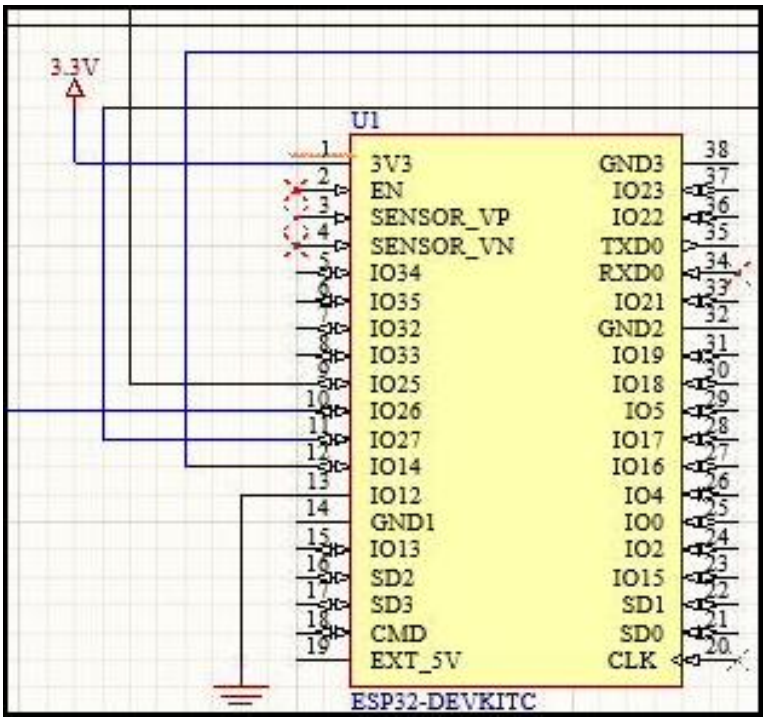


Figure 3.5.S2. ESP32 DevKitC (alarm/gateway board) GPIO assignment.

Results

Both boards are schematic-complete with GPIO assignments settled, etched, and drilled. ESP-NOW pairing and the gateway's external feed polling have not been written yet, so the two boards cannot currently communicate with each other.

Future Progress

- Write and test ESP-NOW pairing between the two boards using real feature packets once Subsystem 3.4's FFT pipeline is merged into the live firmware.
- Implement the gateway's EMSC LastQuake and USGS GeoJSON polling and the Stage 1 to Stage 2 validation logic.

Conclusion

The hardware side of this subsystem is done for both boards. What is left is entirely firmware: getting the two boards actually talking, and getting the gateway checking real external data before it allows a full alarm.

3.6 Alarm and Visual Indicator

Introduction

This subsystem gives the system's decisions a physical presence: three individual through-hole LEDs (red, green, blue) for standby, Stage 1, and Stage 2 status, and a passive piezo buzzer for the audible Stage 2 alarm. All four are driven straight from ESP32 GPIOs with no transistor stage in between, which is a simplification from the original proposal's BJT-buffered design. For this design, instead of one single RGB, three RGB design was implemented where each connected with 220hm resistors.

Component List

Component	Value	Qty	Note
Status LEDs	3 discrete through-hole LEDs, red, green, blue	3	D1, D2, D3
LED series resistors	220 ohm	3	R1, R2, R3, one per LED
Buzzer series resistor	100 ohm	1	R4
Piezo buzzer element	CSS-73B16K-SMT, passive	1	LS1. Capacitive, not inductive, so no flyback diode is needed for direct GPIO drive

Technical Requirements and Constraints

- Direct GPIO drive only works because per-LED current stays well under the ESP32's GPIO sink rating, which the calculation below confirms.
- A passive capacitive buzzer with pieze element was used to keep the direct GPIO safe. A flyback diode and transistor would have been needed if an inductive buzzer had been used.
- To produce sound from a passive piezo element, the buzzer needs a PWM and not static logic level.

Design Justification: LED Channel Current

Given	Vbat_max = 4.2V (fresh LiPo); Vf_red is about 1.8V for a typical red LED, which is the worst case for highest current; R = 220 ohm
Formula	I = (Vbat - Vf) / R
Substitution	I = (4.2 - 1.8) / 220
Result	I = 10.9mA

Gantt Chart

Figure 3.6.G tracks Subsystem 3.6's build schedule from project assignment (30 Jun 2026) to this report's submission (01 Sep 2026). Design, PCB layout, and breadboard test of the LED and buzzer drive are all complete.

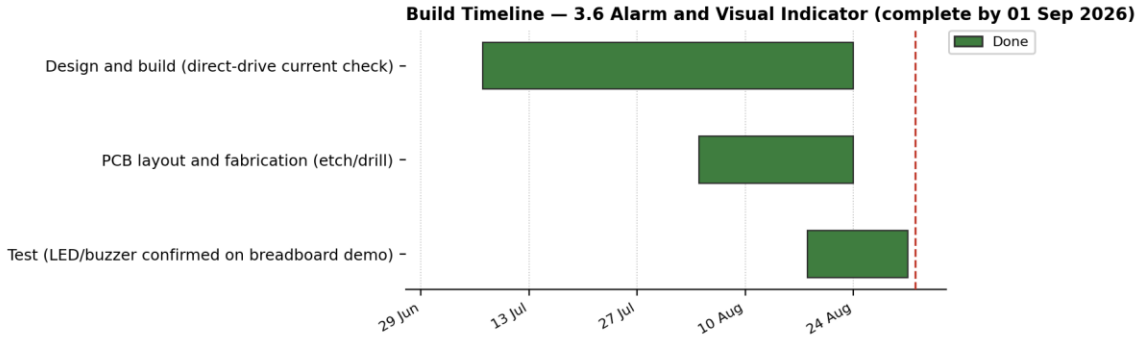


Figure 3.6.G. Subsystem 3.6 build timeline. All phases shown were completed by the report submission date.

Block Diagram

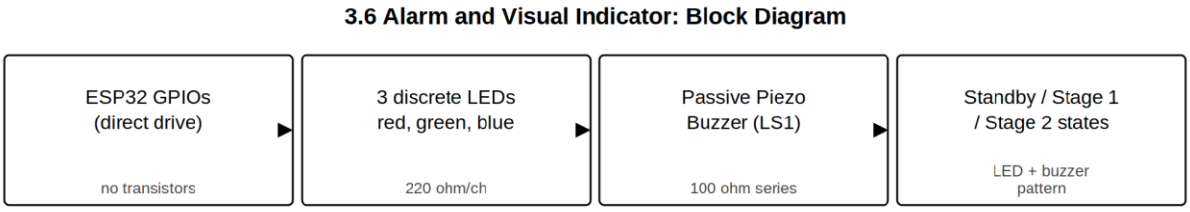


Figure 3.6.0. The LEDs and buzzer resistors are directly driven by GPIO.

Methodology

Three LEDs are synced using three GPIOs and 220 ohm resistors and another GPIO drives the buzzer through a 100 ohm resistor using PWM. At the end of this report, the full schematic structure of the alarm circuit using ESP32, TP4056, MPU6050, resistors, LEDs and Buzzers is shown.

Schematic

Figure 3.6.S crops the three LEDs and the buzzer, with their series resistors, from the alarm board schematic.

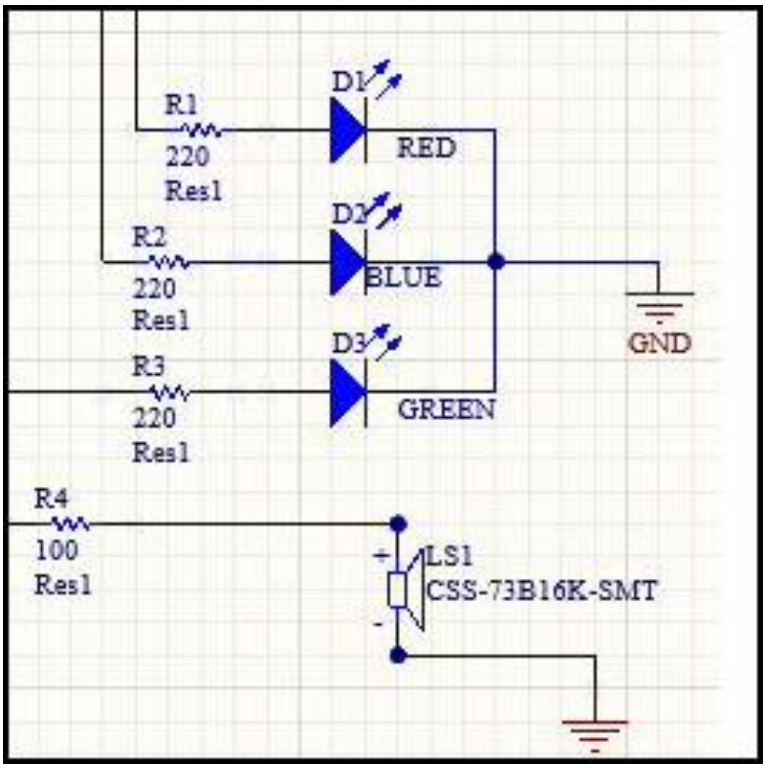


Figure 3.6.S. Three discrete LEDs (R1/D1 red, R2/D2 blue, R3/D3 green) and the buzzer (R4, LS1).

Results

The completed schematic diagram of the GPIO driven PCB has been fabricated, drilled, etched and manufactured proving the designing process to be accurate however no components have been soldered down yet on the PCB board.

Future Progress

- Implement the alarm board on PCB and cross check the outputs with the calculated data.
- Once the 3.2 and 3.4 logic are finalized and implemented on hardware, the stage 1 and stage 2 output needs to be verified.

Conclusion

The dropping of the transistor included design made this simple and it was easy to etch and build without compromising safety and it being manufactured is a proof that the layout was indeed correct and buildable. The only task left to do is wiring it on the finalized detection method of 3.2 and 3.4.

7. Hardware Implementation

The system up to this point describes everything that is designed in Altium and validated in Proteus and MATLAB.But everything we validated and did cannot be implemented in hardware.Despite some limitations we got the hardware fully working.

7.1 Hardware Differences with Simulation

	Planned (design)	**Actual (breadboard)**
Boards	Two separate PCBs, Sensing and Alarm, each with its own ESP32	A single ESP32 runs sensing, consensus, and alarm output together
Link	ESP-NOW carries only extracted FFT features between the two nodes	Two circuits were built to test the node system but the Alarm Circuit was integrated
Validation	A gateway node cross-checks EMSC and USGS before sounding the alarm	Not yet reachable, since there is no second node to act as a gateway

Building the full signal chain, sensing, filtering, consensus logic, and alarm output, on a single breadboard rather than splitting it across two PCBs from the start was a deliberate simplification, not an oversight.

7.2 Detection Method: FFT Designed, RMS Deployed

	Planned (design)	**Actual (breadboard)**
Method	256-point FFT with a Hamming window, 1 to 10Hz band energy as the core discriminator	RMS and amplitude thresholding, with detection weight shifted toward the MPU-6050

Why the piezo channel needed rebalancing: According to the rate of deformation, piezoelectric disc generates charge and that is why a sharp tap gives a better signal but slow shaking like earthquake stays in flat baseline. The breadboard implementation data shows that the piezoelectric sensor during 5 seconds of gradual shake stayed at 142.00mV resting value, however the accelerometer output climbed from 10.05 to 13.10 m/s^2 over the same window. This showcases that without using both the sensors, the chances of missing a signal was high.

What changed: Since MPU6050 detecs acceleration in any direction whether it is a sharp movement ot slow gradual movement, which is why it was chosen as a primary sensor. The piezo electric sensor is kept as a secondary option for sharp taps only. It shifts the focus that the FFT value from 3.3 and 3.4 are deciding factors for triggering the alarm circuit. Merging both FFT and the dual sensor verification remains a part for the future work.

7.3 Power: Direct Supply Instead of Battery

The schematics for both boards include a full TP4056 charging path from a LiPo cell. The current hardware demonstration, shown running end to end in Figure 7.1, instead feeds the LM2596 buck stage directly from a regulated external supply over USB. Subsystem 3.1 covers the reasoning and the consequences in detail: running from USB means the design's real LM2596 headroom problem has not yet been encountered on the bench, since USB's roughly 5V clears the 4.8V minimum the buck module needs, while a single-cell LiPo's 3.0 to 4.2V range does not.

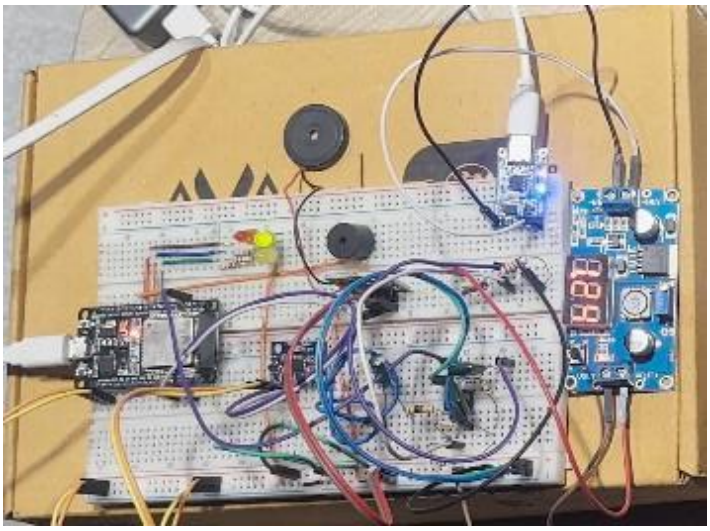


Figure 7.1. The working breadboard demonstration: MPU-6050, piezo module, LM358N filter stage, RGB LED, buzzer, and ESP32, all powered from a USB-fed buck module rather than a battery.

7.4 What This Means Going Forward

- The detection logic, consensus checking, and alarm output are proven on real hardware, end to end, which is a stronger result than a design that only exists in simulation.
- Splitting the working single-board firmware back across two physical nodes, and getting ESP-NOW working between them, is the next concrete milestone, not a redesign.
- The FFT-versus-RMS question is not really open anymore in the sense of which is correct; the RMS approach with MPU-6050-weighted consensus is what the data supports. The remaining work is deciding whether to merge FFT band energy in as a second layer of frequency-domain rejection on top of it, or treat the two as genuinely separate, redundant checks.
- Battery operation needs the LM2596 headroom question resolved first. Testing it as-is on a real LiPo cell without addressing that would just reproduce the sag the calculation in Subsystem 3.1 predicts.

8. Subsystem Timelines

The subsystem-level design, build, and test phases referenced from each subsystem description are collected here as a single reference table, condensed to two phases per subsystem, design and build together, then test, since for a breadboard-first project the two happen together rather than in strict sequence. Dates follow the same draft schedule as the master timeline in Section 5, and status reflects what the working hardware demonstration in Section 7 has actually confirmed, not just what has been drawn in Altium.

Subsystem	**Phase**	**Window**	**Status**
3.1 Primary Power Regulation	Design and build (schematic capture, etched boards)	Wk 1-8	Done
	Test (direct-power bring-up done; battery/LM2596 headroom test still open)	Wk 8-11	In progress
3.2 Dual-Axis Seismic Sensing	Design and build (sensor wiring, calibration firmware)	Wk 1-7	Done
	Test (calibration and consensus confirmed on breadboard)	Wk 7-9	Done
3.3 Active Hardware Filter	Design and build (bias network, Sallen-Key sizing)	Wk 2-6	Done
	Test (Proteus, earthquake/door slam/train, front-end gain topology open)	Wk 6-11	In progress
3.4 FFT-Based Signal Processing	Design and build (sample rate, MATLAB cross-check)	Wk 3-10	Done
	Test (validated in simulation; not yet merged into live firmware)	Wk 10-13	Planned

Subsystem	**Phase**	**Window**	**Status**
3.5 Microcontroller and IoT Communication	Design and build (schematic capture, both boards)	Wk 1-8	Done
	Test (ESP-NOW pairing and gateway validation, single-board today)	Wk 10-13	Planned
3.6 Alarm and Visual Indicator	Design and build (direct-drive current check, PCB layout)	Wk 2-8	Done
	Test (LED and buzzer response confirmed on breadboard demo)	Wk 8-9	Done

9. Schematics

The Sensing PCB carries Subsystems 3.1, 3.2, 3.3, and 3.5; the Alarm/Gateway PCB carries the corresponding parts of Subsystems 3.1, 3.5, and 3.6. Per-subsystem schematic crops are shown alongside each subsystem's own description in Section 3, under that subsystem's "Schematic" heading; this section keeps only the full-board sheets.

9.1 Full Board Schematics

Figure 9.1 is the complete sensing-board schematic: the TP4056 and LM2596 power stage, the MPU-6050 and piezo sensing inputs, the LM358N filter stage, and the ESP32-S3 DevKitC, all on one Altium sheet.

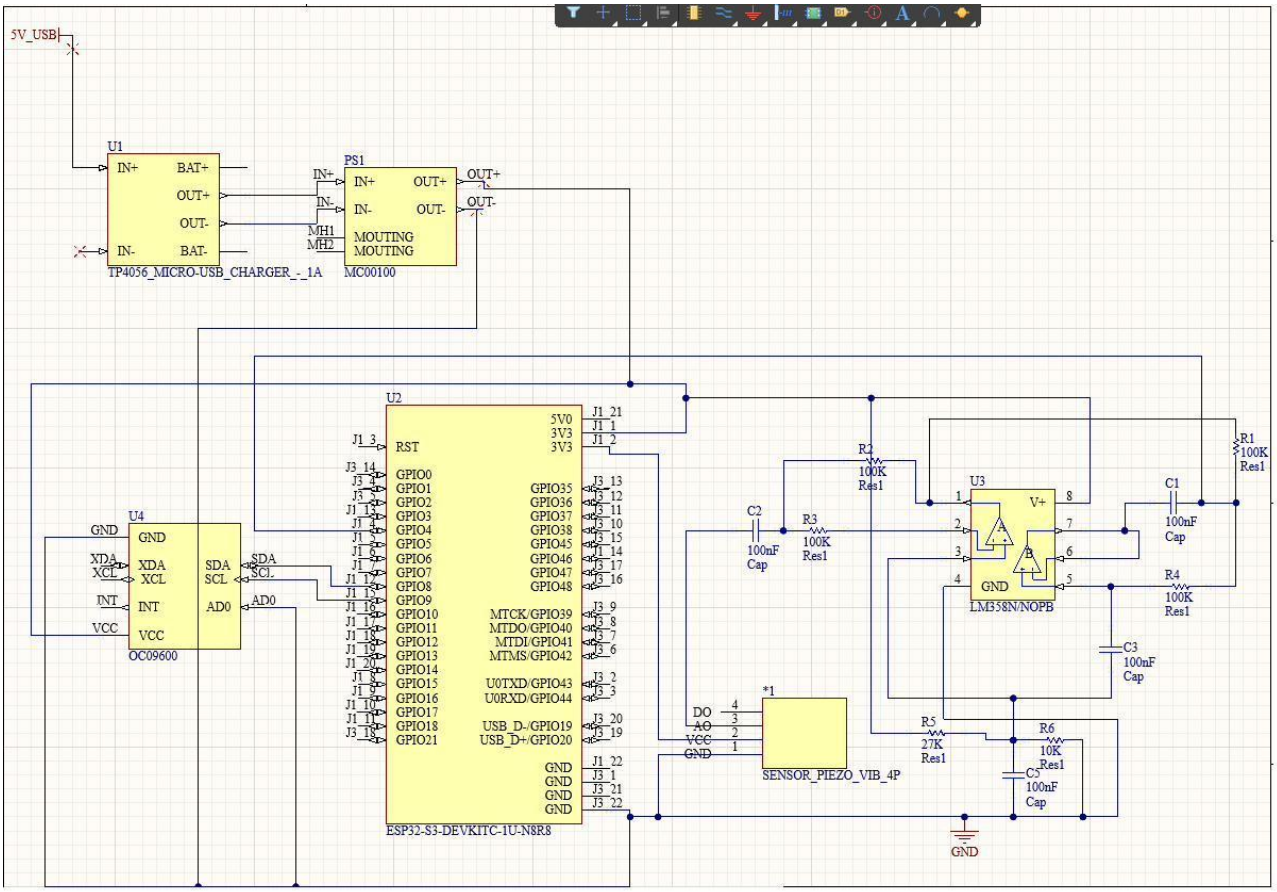


Figure 9.1. Sensing PCB schematic. TP4056 (U1) feeds the LM2596 module (PS1), which powers the ESP32-S3 DevKitC (U2), the MPU-6050 breakout (U4), and the LM358N filter stage (U3).

Figure 9.2 is the complete alarm/gateway-board schematic: its own TP4056 and LM2596 power stage, the ESP32 DevKitC, and the three LEDs and buzzer.

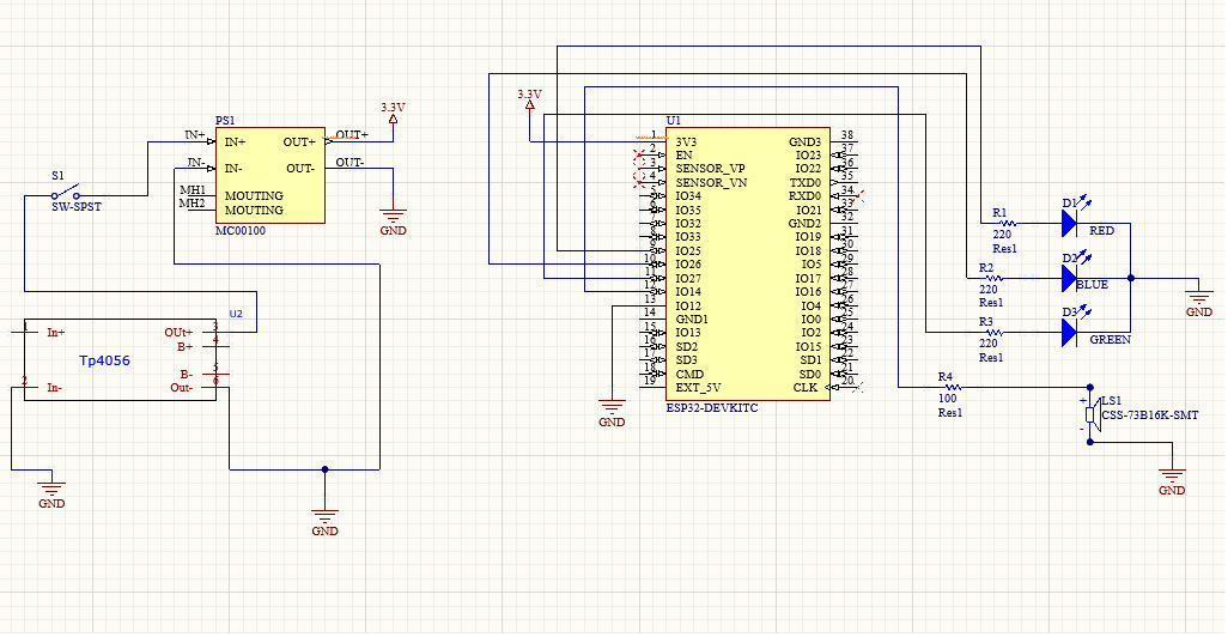


Figure 9.2. Alarm/gateway board schematic. TP4056 (U2) and LM2596 (PS1) power the ESP32 DevKitC (U1), which drives the three LEDs and the buzzer directly.

Figure 9.3 is the Proteus test schematic used for the filter and FFT validation work in Subsystems 3.3 and 3.4, with an Arduino Mega 2560 standing in for the ESP32 and four synthetic noise generators feeding the same filter chain as the real earthquake signal.

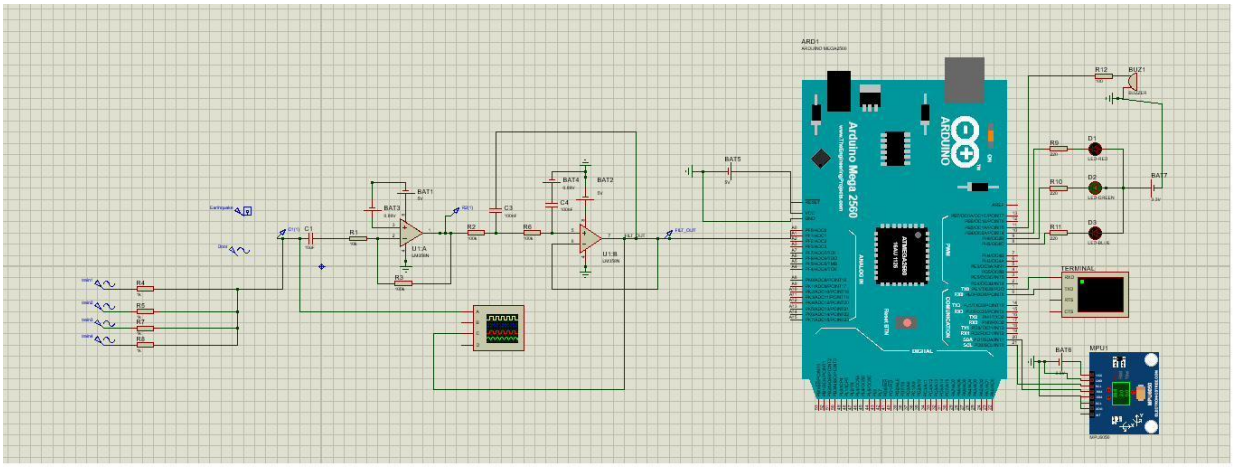


Figure 9.3. Proteus simulation schematic used for filter and FFT validation.

10. PCB Layouts

Both boards are two-layer, etched and drilled at home from toner-transferred artwork, and checked through an online fabrication tool's design-for-manufacturing report. Neither has components populated at the time of this report.

10.1 Etched Boards

Figure 10.1 shows both boards after etching and drilling, before component population: the larger board is the Sensing PCB, the smaller is the Alarm/Gateway PCB.

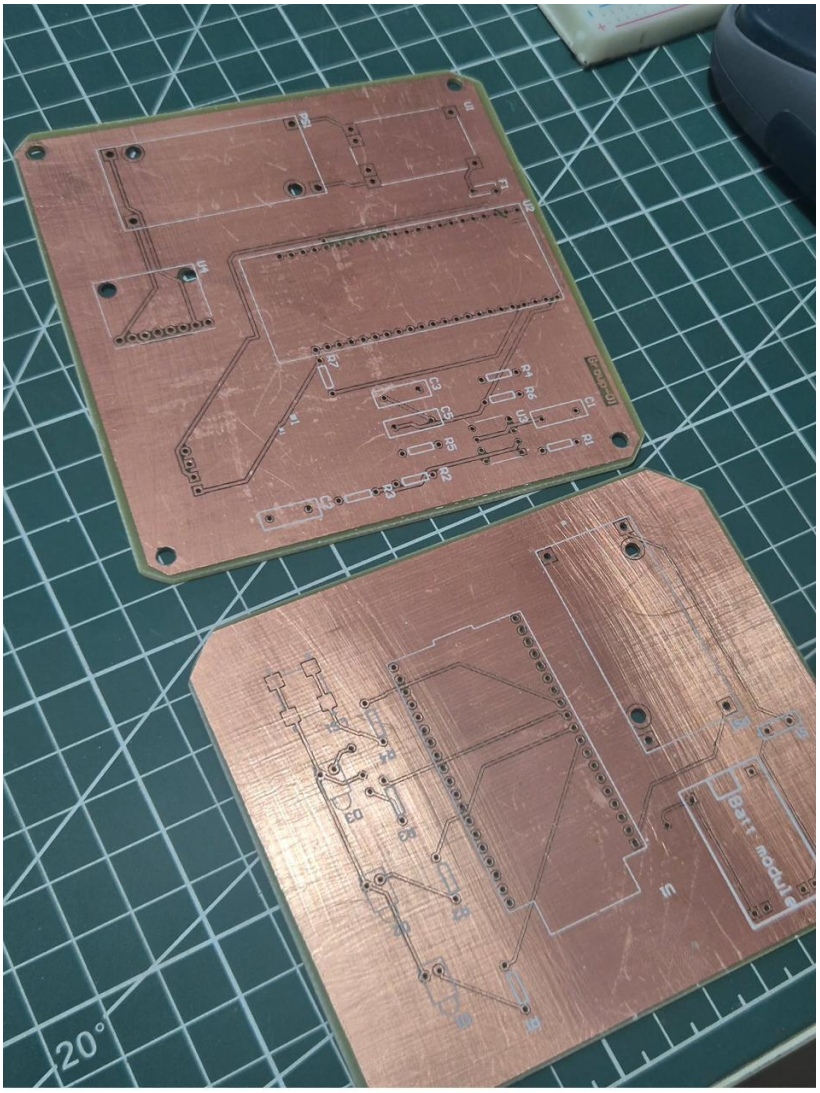
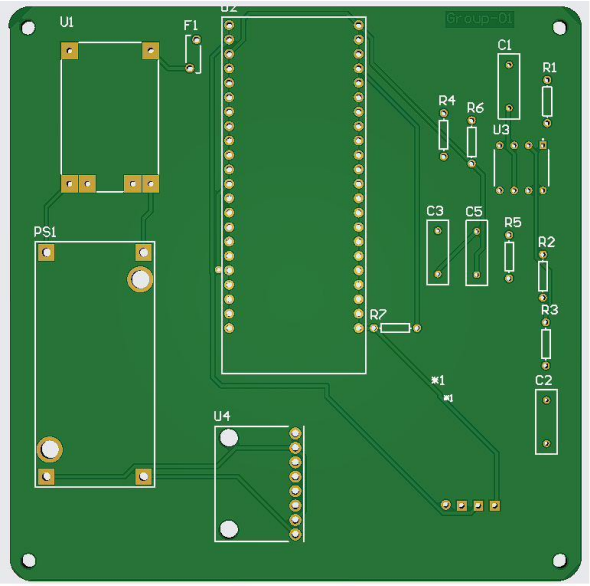


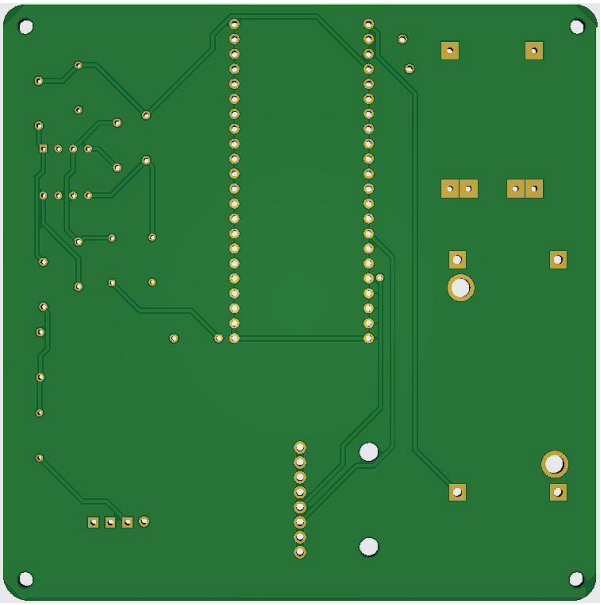
Figure 10.1. Etched and drilled copper for the Sensing PCB (top) and Alarm/Gateway PCB (bottom), unpopulated.

10.2 Sensing PCB

Figure 10.2 shows the Sensing PCB's Altium layout: top layer, bottom layer, and a 3D render with the LM2596 module, ESP32-S3 DevKitC, LM358N, and MPU-6050 breakout placed. This board carries Subsystems 3.1, 3.2, 3.3, and the sensing-side half of 3.5.



Top layer



Bottom layer

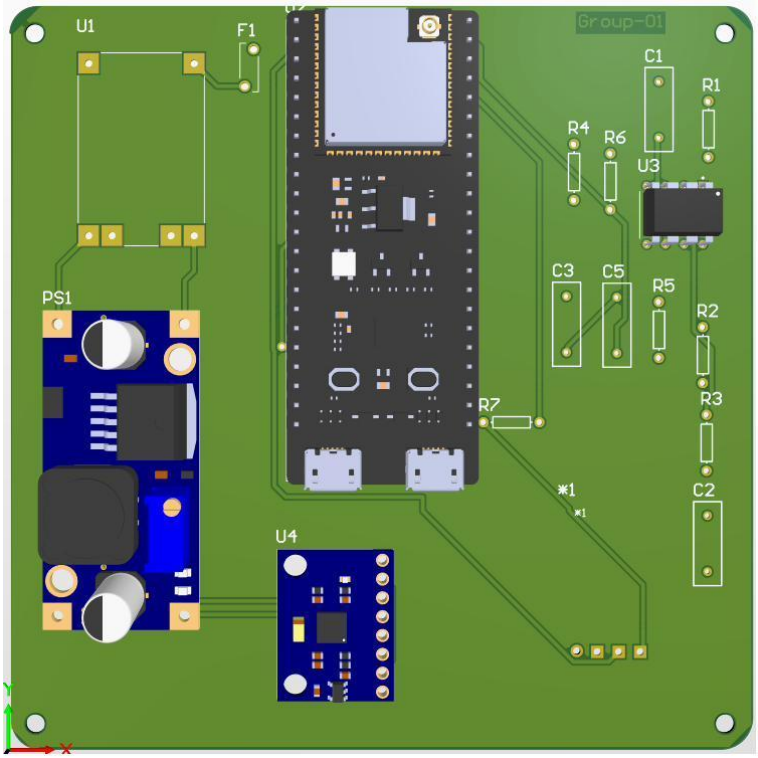
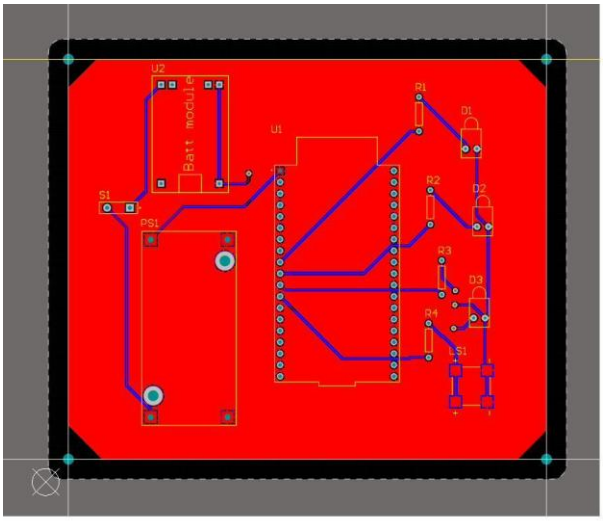


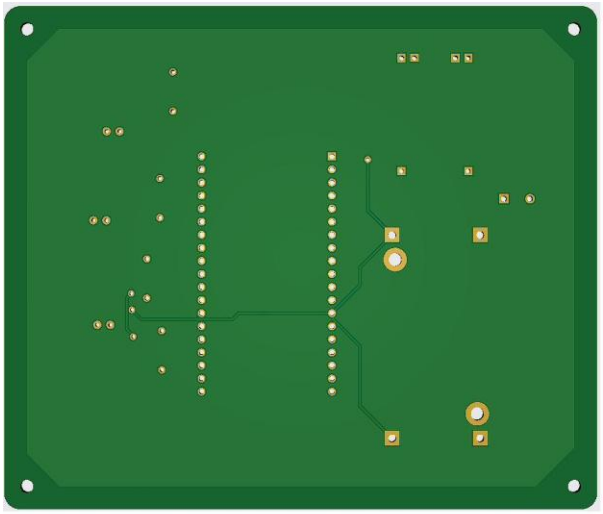
Figure 10.2. Sensing PCB, top layer, bottom layer, and 3D render.

10.3 Alarm and Gateway PCB

Figure 10.3 shows the Alarm/Gateway PCB's Altium layout in the same three views. This board carries the alarm-side half of Subsystem 3.5 and all of Subsystem 3.6.



Top Layer



Bottom layer

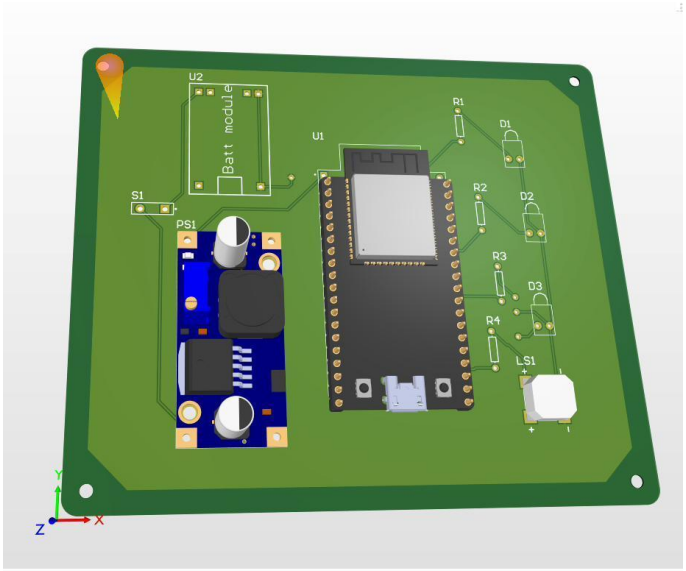


Figure 10.3. Alarm/Gateway PCB, top copper, bottom copper, and 3D render, with the ESP32 DevKitC, LM2596 module, and LED/buzzer footprints placed.

11. PCB Specifications

This section pulls together the physical board specifications, board outline dimensions and the routing width rule, that back up the layout and trace-sizing claims made throughout the Subsystem Descriptions. Both figures come directly from the Altium project rather than being re-derived, dimensions from the Guide Manager's placed reference guides, and trace width from the board's own routing constraint rule.

11.1 Board Outline Dimensions

Each board's outline is set by a pair of horizontal and a pair of vertical guides in Altium's Guide Manager. The distance between each pair gives the outline size directly.

Board	Horizontal guides (Y)	Vertical guides (X)	Outline size
Sensing PCB	200mil, 3700mil	200mil, 4400mil	4200mil x 3500mil (4.2in x 3.5in, about 107mm x 89mm)
Alarm/Gateway PCB	0mil, 4000mil	0mil, 4000mil	4000mil x 4000mil (4.0in x 4.0in, about 102mm x 102mm)

11.2 Routing Width Rule

The 20mil trace width used in the Subsystem 3.1 IPC-2221 calculation is not just a hand assumption. It is set as an actual Altium design rule, with minimum, preferred, and maximum width all pinned to 20mil on both copper layers, so the autorouter and the manual routing are both held to the same constraint the calculation assumes.

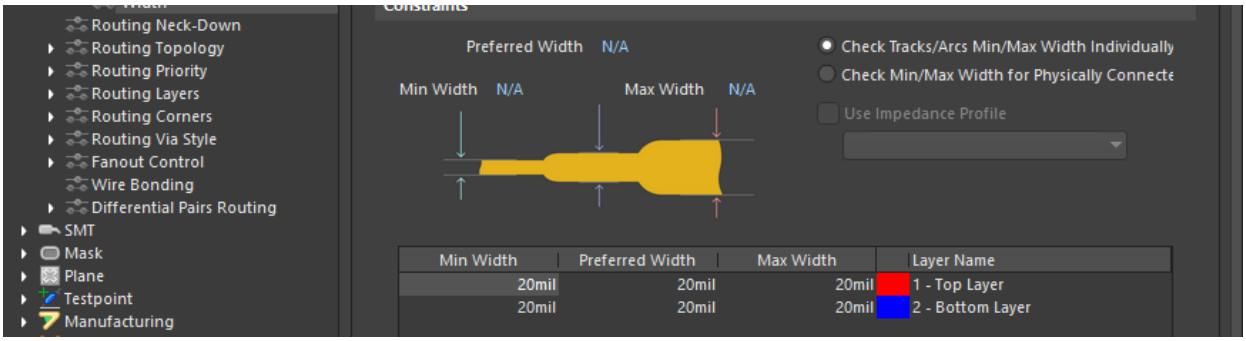


Figure 11.1. Altium routing width constraint: 20mil minimum, preferred, and maximum on the Top Layer and Bottom Layer.

Layer	**Min width**	**Preferred width**	**Max width**
1, Top Layer	20mil	20mil	20mil
2, Bottom Layer	20mil	20mil	20mil

Appendix A. Original Procurement Bill of Materials

This is the itemized, costed Bill of Materials from the original project proposal, sourced primarily from RoboticsBD with TechShopBD and the local Nawabpur Road/Stadium Market as fallbacks. It documents the initial single-board concept: one ESP32-WROOM-32, an OLED status display, a common-cathode RGB LED, and BC547/2N2222 transistor drivers for the outputs.

The design evolved after this costing was done, into the two-PCB architecture described in Section 4: the OLED display was dropped, the RGB LED became three discrete through-hole LEDs, the transistor drivers were removed in favor of driving the LEDs and buzzer directly from ESP32 GPIOs, and the single ESP32-WROOM-32 became an ESP32-S3 on the Sensing PCB paired with a classic ESP32 on the Alarm/Gateway PCB. Section 4 is the current, as-built component list; the table below is kept here as the original sourcing and cost record.

Item	Part	Notes	Qty	Unit (৳)	Total (৳)	Source
Microcontroller	ESP32-WROOM-32	Central processor: consensus logic, PWM alerts, Wi-Fi API polling	1	720	720	RoboticsBD
Motion Sensor	MPU-6050	3-axis accelerometer/gyro (I2C), one per satellite node	2	372	744	RoboticsBD
Shock Detector	Piezo Vibration Sensor Module	Analog vibration transducer disc, one per node	2	180	360	RoboticsBD
Signal Conditioning	LM358DR Op-Amp	Dual op-amp (SOIC-8), 2nd-order Sallen-Key LPF, fc=15 Hz	2	20	40	RoboticsBD
Display Module	0.96" OLED (I2C)	128x64 display, live acceleration/status readout	1	395	395	RoboticsBD
Visual Alert	RGB LED (Common Cathode)	3-state PWM status indicator	1	20	20	RoboticsBD
Audio Alert	Passive Buzzer	Acoustic alarm, driven via BC547 switch	1	15	15	RoboticsBD
Transistor Drivers	2N2222 / BC547	NPN BJTs: 3x RGB lines + 1x buzzer	4	5	20	RoboticsBD

Item	Part	Notes	Qty	Unit (৳)	Total (৳)	Source
Power Reg. (LDO + Charger)	AMS1117-3.3, 3.7V LiPo, TP4056	Linear 3.3V regulator, battery, USB charger	1	652	652	RoboticsBD
Power Reg. (Buck)	LM2596	Buck converter, 7-12V to 5V	1	85	85	RoboticsBD
Overcurrent Protection	Resettable Polyfuse	PTC fuse, 500 mA	1	10	10	TechShopBD
Polarity Protection	1N5819 Schottky Diode	Reverse-polarity protection	1	5	5	RoboticsBD
Signal Diodes	1N4148	Flyback + clamp diodes	5	5	25	RoboticsBD
Connectors	JST-XH 6-Pin Headers	Board-to-board detachable headers	4	30	120	RoboticsBD
Passives & Discretes	Resistors & Capacitors	Assorted 0805 passives	1	400	400	TechShopBD
Miscellaneous	Resistors, caps, wires, jumpers	General assembly	var	200	300	Local market
					Grand Total: ৳ 3,911	

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